Jet flavour tagging

UnifiedParticleTransformer v2 (UParT)

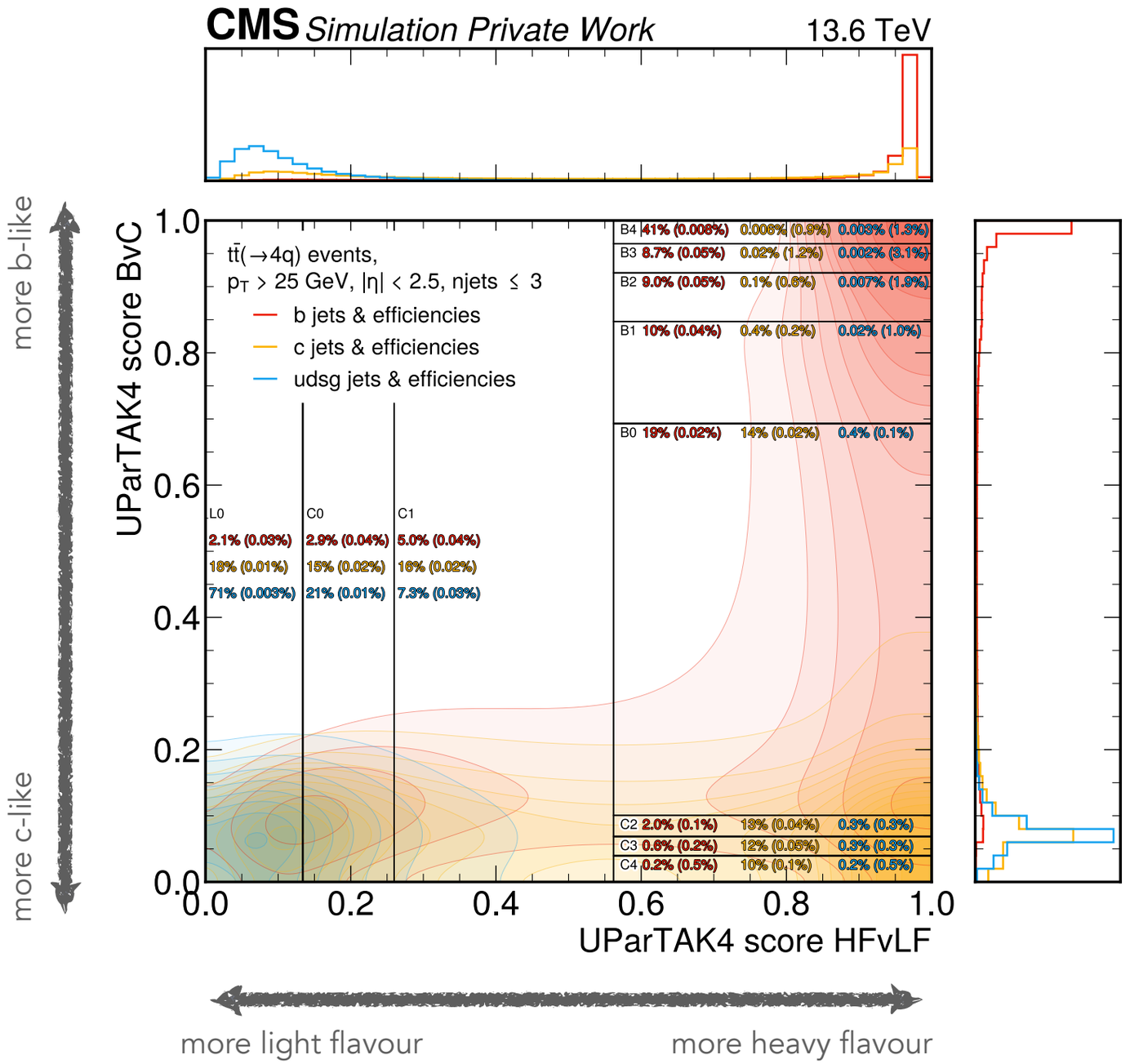
- State-of-the-art jet tagger based on ParticleTransformer
- Standard jet tagging algorithm used in CMS, available in NanoAOD v15

Simultaneous b and c tagging

- Define two discriminators to distinguish heavy vs light flavour jets (HFvLF) and b vs c jets (BvC)

$$\text{HFvLF} = \frac{p(b) + p(bb) + p(c) + p(cc)}{p(b) + p(bb) + p(c) + p(cc) + p(uds) + p(g)}$$

$$\text{BvC} = \frac{p(b) + p(bb)}{p(b) + p(bb) + p(c) + p(cc)}$$



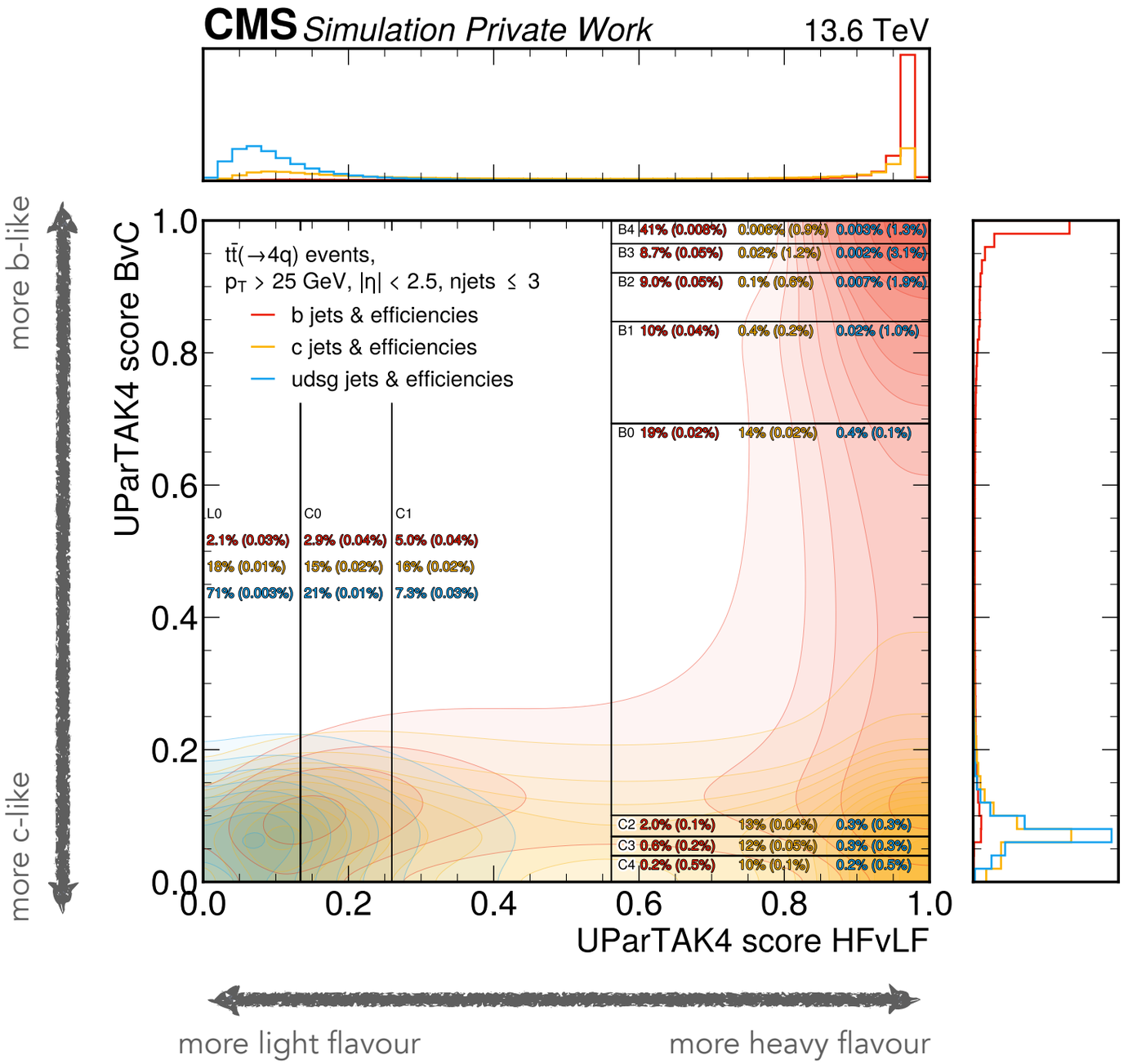
Jet flavour tagging

Custom tagging categories

- **Pseudo-continuous**: use granular score information rather than full discriminant
- More robust calibration than full score
- 11 exclusive tagging categories defined
  - **L0**: light (udsg) jet enriched bin
  - **C0—C4**: c jet enriched bins
  - **B0—B4**: b jet enriched bins

Custom working points

- **Loose (L)**: B0—B4, C0—C4
- **Medium (M)**: B1—B4, C1—C4
- **Tight (T)**: B2—B4, C2—C4



Jet flavour tagging: calibration

Taking existing workflow from HIG-24-018 and developing with BTV

- Use three data samples enriched in different jet flavours: Z+jets (light), W+jets (c), $t\bar{t}$ dileptonic (b)
- Simultaneously fit these regions to derive scale factors (SF) for each of the 11 exclusive tagging categories
  - SFs also derived in bins of jet p_T: [25, 35), [35, 50), [50, 70), [70, 90), [90, 120), [120, inf)
- SFs parametrized to ensure that the original normalization (without tagging requirements) is preserved per truth flavour

